

Paper 4.75 - Boundary Repair as Next-State Precondition

CQE-paper-04.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-04.75.md

Paper 4.75 - Boundary Repair as Next-State Precondition

Purpose

Paper 4.75 explains how boundary repair becomes the precondition for Paper 5's path carrier and for later causal-code review.

Exported State

Paper 4 exports:

```
repaired constraint rows
original LCR states
Paper 3 axis/sheet coordinates
reason: Paper 2 correction fired
status: constraint
next legal routes
source and target paper labels
```

Use in Paper 5

Paper 5 may attach a Paper 4 repair row as a payload only if the payload does not alter the path rule.

The allowed transition is:

```
repair_boundary(s) = constraint row r
attach_payload(q_t, r) = carried receipt at path state q_t
payload_alters_path_rule = false
```

Paper 5 proves the path-carrier rule. Paper 4 supplies the payload object.

Use in Paper 6

Paper 6 may register Paper 4 repair rows as typed causal edges. The edge must name its source, target, edge type, receipt, and status. Paper 4 proves the row; Paper 6 proves the dependency graph discipline.

Use in Later Papers

Later papers may lift boundary repair into:

- rolling path carriers,
- causal proof graphs,
- curvature and geometry repair language,
- mass-residue or Hamiltonian accumulated-center claims,
- product and adapter workflows.

Every lift must state which later theorem closes the added structure.

Precondition Rule

Before a later paper uses Paper 4, it should be able to answer:

```
what failed?
what state and coordinate were preserved?
```

Why is the row a constraint rather than proof?
Which next route consumes it?
Which receipt proves the repair row?

Conclusion

Paper 4.75 turns repair into portable state. It exports failure as structured constraint so
Paper 5 can carry it and Paper 6 can audit it.